

# MATH 20700 Honors Analysis I

## Homework 8

Malhar Manek

### Exercise 1

*Proof.* Let us prove that for any vectors  $A, B, C \in \mathbb{R}^3$ :

$$A \cdot (B \times C) = \det(A, B, C)$$

(In this particular context,  $A = F$ ,  $B = \frac{\partial \varphi}{\partial u}$ ,  $C = \frac{\partial \varphi}{\partial v}$  will complete the specific proof we are asked for).

Let us look at the determinant on the RHS first.

Writing the matrix as columns  $[A, B, C] = \begin{pmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{pmatrix}$ , we can expand this determinant along the first column (the  $A$  column) using cofactor expansion.

This gives us

$$\begin{aligned} \det \begin{pmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{pmatrix} &= A_1 \det \begin{pmatrix} B_2 & C_2 \\ B_3 & C_3 \end{pmatrix} - A_2 \det \begin{pmatrix} B_1 & C_1 \\ B_3 & C_3 \end{pmatrix} + A_3 \det \begin{pmatrix} B_1 & C_1 \\ B_2 & C_2 \end{pmatrix} \\ &= A_1(B_2C_3 - B_3C_2) - A_2(B_1C_3 - B_3C_1) + A_3(B_1C_2 - B_2C_1) \\ &= A \cdot [(B_2C_3 - B_3C_2)\hat{x} - (B_1C_3 - B_3C_1)\hat{y} + (B_1C_2 - B_2C_1)\hat{z}] \\ &= A \cdot (B \times C). \end{aligned}$$

Now for the general case, we set up the  $n \times n$  matrix where the first column is  $F$  and the remaining columns are the tangent vectors  $\frac{\partial \varphi}{\partial u_1}, \dots, \frac{\partial \varphi}{\partial u_{n-1}}$ . We expand this determinant along the first column. The sum looks like:

$$\det \left( F, \frac{\partial \varphi}{\partial u} \right) = \sum_{k=1}^n (-1)^{k+1} F_k \cdot \det(M_{k,1})$$

We observe that the submatrix  $(M_{k,1})$  is exactly the projection of the Jacobian matrix of  $\varphi$  onto the subspace spanned by the basis vectors  $\{e_1, \dots, e_{k-1}, e_{k+1}, \dots, e_n\}$ . This submatrix  $(M_{k,1})$  is the original matrix with the  $F$ -column removed and the  $k$ -th row removed.

The columns of the submatrix  $(M_{k,1})$  are the vectors  $\frac{\partial \varphi}{\partial u}$ . The rows of the submatrix  $(M_{k,1})$  are all coordinates except  $x_k$ .

Hence, the determinant of  $(M_{k,1})$  is exactly the differential form  $dx_1 \wedge \dots \wedge \widehat{dx_k} \wedge \dots \wedge dx_n$  applied to the vectors  $\frac{\partial \varphi}{\partial u}$  (note: the  $\widehat{dx_k}$  notation means “omit this term”).

QED

## Exercise 2

*Proof.* (a) It suffices to show that  $x^2 + y^2 + z^2 = 1$ .

We have

$$\begin{aligned}
 \left( \frac{2u}{1+u^2+v^2} \right)^2 + \left( \frac{2v}{1+u^2+v^2} \right)^2 + \left( \frac{-1+u^2+v^2}{1+u^2+v^2} \right)^2 &= \frac{4u^2 + 4v^2 + (-1+u^2+v^2)^2}{(1+u^2+v^2)^2} \\
 &= \frac{4u^2 + 4v^2 + 1 - 2(u^2+v^2) + (u^2+v^2)^2}{(1+u^2+v^2)^2} \\
 &= \frac{2u^2 + 2v^2 + 1 + (u^2+v^2)^2}{(1+u^2+v^2)^2} \\
 &= \frac{2u^2 + 2v^2 + 1 + (u^2+v^2)^2}{1 + 2(u^2+v^2) + (u^2+v^2)^2} \\
 &= \frac{2u^2 + 2v^2 + 1 + (u^2+v^2)^2}{2u^2 + 2v^2 + 1 + (u^2+v^2)^2} \\
 &= 1.
 \end{aligned}$$

(b) To show that  $\alpha$  is a bijection, let us explicitly construct its inverse.

For simplicity of notation, let us write  $R := u^2 + v^2$ . Then,  $z = \frac{-1+R}{1+R}$ . Then,  $(1+R)z = z + Rz = -1 + R$ , hence  $R - Rz = R(1-z) = z + 1$ , hence  $R = \frac{1+z}{1-z}$ .

Thus,  $1+R = 1 + \frac{1+z}{1-z} = \frac{1-z+1+z}{1-z} = \frac{2}{1-z}$ . Thus,  $x = \frac{2u}{1+R} = \frac{2u}{\frac{2}{1-z}} = u(1-z)$ . Analogously,  $y = v(1-z)$ .

Hence, the inverse function is given by  $u = \frac{x}{1-z}, v = \frac{y}{1-z}$ .

(c)  $dA = \left\| \frac{\partial \alpha}{\partial u} \times \frac{\partial \alpha}{\partial v} \right\|$  so the surface area is

$$\text{Area} = \int dS = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left\| \frac{\partial \alpha}{\partial u} \times \frac{\partial \alpha}{\partial v} \right\| du dv$$

We have

$$\begin{aligned}
 \frac{\partial \alpha}{\partial u} &= \left( \frac{\partial}{\partial u} \left[ \frac{2u}{1+u^2+v^2} \right], \frac{\partial}{\partial u} \left[ \frac{2v}{1+u^2+v^2} \right], \frac{\partial}{\partial u} \left[ 1 - \frac{2}{1+u^2+v^2} \right] \right) \\
 &= \left( \frac{-2u^2 + 2v^2 + 2}{(1+u^2+v^2)^2}, \frac{-4uv}{(1+u^2+v^2)^2}, \frac{4u}{(1+u^2+v^2)^2} \right) \\
 &= \frac{1}{(1+u^2+v^2)^2} \cdot (-2u^2 + 2v^2 + 2, -4uv, 4u)
 \end{aligned}$$

And similarly

$$\begin{aligned}
 \frac{\partial \alpha}{\partial v} &= \left( \frac{\partial}{\partial v} \left[ \frac{2u}{1+u^2+v^2} \right], \frac{\partial}{\partial v} \left[ \frac{2v}{1+u^2+v^2} \right], \frac{\partial}{\partial v} \left[ 1 - \frac{2}{1+u^2+v^2} \right] \right) \\
 &= \left( \frac{-4uv}{(1+u^2+v^2)^2}, \frac{2u^2 - 2v^2 + 2}{(1+u^2+v^2)^2}, \frac{4v}{(1+u^2+v^2)^2} \right) \\
 &= \frac{1}{(1+u^2+v^2)^2} \cdot (-4uv, 2u^2 - 2v^2 + 2, 4v)
 \end{aligned}$$

Let  $A := (-2u^2 + 2v^2 + 2, -4uv, 4u)$ ,  $B := (-4uv, 2u^2 - 2v^2 + 2, 4v)$  and  $k := \frac{1}{(1+u^2+v^2)^2}$  such that  $\frac{\partial \alpha}{\partial u} = kA$ ,  $\frac{\partial \alpha}{\partial v} = kB$ .

We want to find  $kA \times kB = k^2(A \times B)$ . Clearly, by inspection, the dot product  $A \cdot B$  is zero, hence the vectors  $A$  and  $B$  are orthogonal, hence  $\|A \times B\| = \|A\| \cdot \|B\|$ .

$$\|A\| = A \cdot A = 2(1 + u^2 + v^2) = B \cdot B = \|B\|.$$

$$\begin{aligned} \text{Area} &= \int dS = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} k^2 \|A\| \cdot \|B\| \, du \, dv \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{4(1 + u^2 + v^2)^2}{(1 + u^2 + v^2)^4} \, du \, dv \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{4}{(1 + u^2 + v^2)^2} \, du \, dv \\ &= \int_0^{2\pi} \int_0^{\infty} \frac{4}{(1 + r^2)^2} r \, dr \, d\theta \\ &= 2\pi \int_0^{\infty} \frac{4}{(1 + r^2)^2} r \, dr \\ &= 4\pi \int_1^{\infty} \frac{1}{w^2} \, dw \\ &= 4\pi. \end{aligned}$$

(We use the substitution  $w = 1 + r^2$  above).

(d)

QED

### Exercise 3

*Proof.* (a) Note that  $\mathbf{n} = \frac{\mathbf{r}}{|\mathbf{r}|}$

Since  $F = \frac{\mathbf{r}}{|\mathbf{r}|^3}$ , we get that the dot product  $\mathbf{F} \cdot \mathbf{n} = \frac{\mathbf{r}}{|\mathbf{r}|^3} \cdot \frac{\mathbf{r}}{|\mathbf{r}|} = \frac{1}{|\mathbf{r}|^2}$

Thus,  $\iint_{S_r} \frac{1}{|\mathbf{r}|^2} = \frac{1}{r^2} \iint_{S_r} dA = \frac{1}{r^2} (4\pi r^2) = 4\pi$ .

(b)

$$\frac{\partial}{\partial x} \left( \frac{x}{|\mathbf{r}|^3} \right) = \frac{1}{(x^2 + y^2 + z^2)^{3/2}} - \frac{3x^2}{(x^2 + y^2 + z^2)^{5/2}}$$

$$\frac{\partial}{\partial y} \left( \frac{y}{|\mathbf{r}|^3} \right) = \frac{1}{(x^2 + y^2 + z^2)^{3/2}} - \frac{3y^2}{(x^2 + y^2 + z^2)^{5/2}}$$

$$\frac{\partial}{\partial z} \left( \frac{z}{|\mathbf{r}|^3} \right) = \frac{1}{(x^2 + y^2 + z^2)^{3/2}} - \frac{3z^2}{(x^2 + y^2 + z^2)^{5/2}}$$

Hence

$$\operatorname{div} F = \frac{3}{(x^2 + y^2 + z^2)^{3/2}} - \frac{3(x^2 + y^2 + z^2)}{(x^2 + y^2 + z^2)^{5/2}} = \frac{3}{(x^2 + y^2 + z^2)^{3/2}} - \frac{3}{(x^2 + y^2 + z^2)^{3/2}} = 0$$

(c) Since the divergence is undefined at the origin, let us create a new volume  $V = \Omega \setminus B_\epsilon(\mathbf{0})$ .

Then,  $\partial V = \partial\Omega \cup S_\epsilon(\mathbf{0})$ .

By the divergence theorem,

$$\iiint_{\Omega \setminus B_\epsilon(\mathbf{0})} \operatorname{div} F \, dV = 0 \text{ implies } \iint_{\partial\Omega} \mathbf{F} \cdot \mathbf{n}_{\partial\Omega} \, dA + \iint_{S_\epsilon} \mathbf{F} \cdot \mathbf{n}_{S_\epsilon} \, dA = 0$$

Observe that on the inner surface  $S_\epsilon$ , the normal vector points towards the origin. Hence,

$$\iint_{S_\epsilon} \mathbf{F} \cdot \mathbf{n}_{S_\epsilon} \, dA = -4\pi.$$

$$\text{Hence, } \iint_{\partial\Omega} \mathbf{F} \cdot \mathbf{n}_{\partial\Omega} \, dA = 4\pi.$$

QED

## Exercise 4

*Proof.* We first find the general product rule  $\nabla \cdot (f\mathbf{A}) = \frac{\partial f}{\partial x}\mathbf{A}_x + \frac{\partial f}{\partial y}\mathbf{A}_y + f(\nabla \cdot \mathbf{A}) = (\nabla f \cdot \mathbf{A}) + f(\nabla \cdot \mathbf{A})$

1. Thus,  $|\nabla u|^2 + u\Delta u = (\nabla u \cdot \nabla u) + u(\nabla \cdot (\nabla u)) = \nabla \cdot (u\nabla u) = \operatorname{div} (u\nabla u)$ .

Above we use the substitution  $f = u$ ,  $\mathbf{A} = \nabla u$ .

$$\text{Hence, by divergence theorem, we get } \int_{\Omega} |\nabla u|^2 + u\Delta u \, dx = \int_{\Omega} \operatorname{div} (u\nabla u) \, dx = \int_{\partial\Omega} (u\nabla u) \cdot \mathbf{n} \, dA$$

$$\text{We now want to show that } \int_{\partial\Omega} (u\nabla u) \cdot \mathbf{n} \, dA = \int_{\partial\Omega} u \partial_n u \, dA$$

Intuitively, it is clear that  $\partial_n u = \nabla u \cdot \mathbf{n}$ , i.e., the component of the gradient  $\nabla u$  along the direction  $\mathbf{n}$ . Hence,  $u \partial_n u = u(\nabla u \cdot \mathbf{n}) = (u\nabla u) \cdot \mathbf{n}$ . This proves the first identity.

2. Setting  $f = u$  and  $\mathbf{A} = \nabla v$  in the general product rule, we get  $\nabla \cdot (u\nabla v) = (\nabla u \cdot \nabla v) + u(\nabla \cdot \nabla v) = (\nabla u \cdot \nabla v) + u\Delta v$ .

Now setting  $f = v$  and  $\mathbf{A} = \nabla u$  in the general product rule, we get  $\nabla \cdot (v\nabla u) = (\nabla v \cdot \nabla u) + v(\nabla \cdot \nabla u) = (\nabla v \cdot \nabla u) + v\Delta u$ .

Subtracting, we get  $\nabla \cdot (u\nabla v) - \nabla \cdot (v\nabla u) = (\nabla u \cdot \nabla v) + u\Delta v - (\nabla v \cdot \nabla u) - v\Delta u = u\Delta v - v\Delta u = \nabla \cdot (u\nabla v - v\nabla u)$ .

$$\text{Hence, } \int_{\Omega} u\Delta v - v\Delta u \, dx = \int_{\Omega} \nabla \cdot (u\nabla v - v\nabla u) \, dx = \int_{\partial\Omega} (u\nabla v - v\nabla u) \cdot \mathbf{n} \, dA = \int_{\partial\Omega} u \partial_n v - v \partial_n u \, dA.$$

QED

## Exercise 5

*Proof.* Let  $\Omega = \mathbb{R}^2 \setminus \{P_1\}$  for some point  $P_1 \in \mathbb{R}^2$ .

Consider the angle  $\theta$  in polar coordinates. Let  $F = \nabla\theta$ . Clearly  $\text{curl } F = 0$  by Exercise 7 (d).

Since  $\int_a^b \nabla f \cdot ds = f(b) - f(a)$ , we have  $\oint \nabla\theta \cdot ds = 2\pi$ . And if you wind around that hole  $k$  times, the integral becomes  $2\pi k$ . So for a domain with one hole, the set of all possible line integrals is  $\{2\pi k \mid k \in \mathbb{Z}\}$  (which is just  $0, \pm 2\pi, \pm 4\pi, \dots$ ). Clearly, this set is not dense in  $\mathbb{R}$ .

But now let us consider a similar construction with two holes, such that  $\Omega = \mathbb{R}^2 \setminus \{P_1 \cup P_2\}$ . Define  $F = \alpha \nabla\theta_1 + \beta \nabla\theta_2$  with  $\alpha, \beta \in \mathbb{R}$ .

If we consider a loop  $C$  that winds  $n$  times around  $P_1$  and  $m$  times around  $P_2$  (where  $n, m$  are integers), then the integral  $\oint_C F \cdot \tau ds = 2\pi\alpha n + 2\pi\beta m = 2\pi(\alpha n + \beta m)$ .

We want the set of all possible values  $\{2\pi(\alpha n + \beta m) \mid n, m \in \mathbb{Z}\}$  to be dense in  $\mathbb{R}$ , so pick  $\alpha, \beta \in \mathbb{R}$  such that  $\frac{\alpha}{\beta}$  is irrational.

Hence we define  $F = \sqrt{2}\nabla\theta_1 + \nabla\theta_2$ .

Note: if we want Cartesian coordinates, we can pick holes at  $(0, 1)$  and  $(0, 0)$  (i.e.,  $\Omega = \mathbb{R}^2 \setminus \{(0, 1) \cup (0, 0)\}$ ) and we define  $F = \sqrt{2} \frac{1}{x^2+y^2} \begin{pmatrix} -y \\ x \end{pmatrix} + \frac{1}{(x-1)^2+y^2} \begin{pmatrix} -y \\ x-1 \end{pmatrix}$ .

QED

## Exercise 6

*Proof.* 1. Let us write  $F = (P, Q)$ , then  $F^\perp = (-Q, P)$ .

By Green's theorem, we have  $0 = \oint_C F^\perp \cdot \tau ds = \iint \left( \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dA$ .

Hence, the integrand  $\left( \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) = \text{div } F = 0$ . This is because  $F$  is  $C^1$  hence  $\text{div } F$  is continuous.

2. Suppose two solution curves, let us call them  $C_1$  and  $C_2$ , start at the same point  $P$  and then split to follow different paths.

We observe that for any solution curve, the tangent vector  $\tau$  points in the same direction as the vector field  $F$ .

If you calculate the line integral  $\int F^\perp \cdot \tau ds$  along strictly one of these solution trajectories (not a closed loop yet, just a segment of the solution), the value of the integral is 0 because  $F^\perp \cdot \tau = 0$ .

Imagine the two solution curves split at a point  $P$ . Consider a small “wedge-shaped” region bounded by three sides:

- (1) A segment of the first solution curve starting from  $P$ .
- (2) A segment of the second solution curve starting from  $P$ .

- (3) A short straight line segment  $L$  connecting the endpoints of those two segments.

This is a closed loop, hence the integral over this closed loop must be zero (since the divergence of  $F$  is zero). Since the integral over any segment of a solution curve is zero, we observe that parts (1) and (2) contribute zero to the integral. Hence, part (3) must have integral equal to zero.

Since  $L$  connects separating curves, intuitively we see that it cuts across the flow. This means  $L$  is roughly parallel to  $F^\perp$ . Since  $F \neq 0$ , we know that  $F^\perp \neq 0$ , hence the dot product  $F^\perp \cdot \tau$  along  $L$  is strictly non-zero (it has a definite sign). Integrating a non-zero signed quantity gives a non-zero result.

This is a contradiction.

QED

## Exercise 7

*Proof.* (a)

$$\begin{aligned} d\omega_F &= \left( \frac{\partial F_1}{\partial x} dx + \frac{\partial F_1}{\partial y} dy + \frac{\partial F_1}{\partial z} dz \right) dx + \left( \frac{\partial F_2}{\partial x} dx + \frac{\partial F_2}{\partial y} dy + \frac{\partial F_2}{\partial z} dz \right) dy \\ &\quad + \left( \frac{\partial F_3}{\partial x} dx + \frac{\partial F_3}{\partial y} dy + \frac{\partial F_3}{\partial z} dz \right) dz \\ &= \left( \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dx dy + \left( \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) dy dz + \left( \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) dz dx \end{aligned}$$

We want to find  $G$  such that  $d\omega_F = \omega_G^* = G_1 dy dz + G_2 dz dx + G_3 dx dy$ .

Hence,  $G$  is given by  $G_1 = \left( \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right)$ ,  $G_2 = \left( \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right)$ ,  $G_3 = \left( \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right)$ .

(b)  $dp = \frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz$ .

The gradient vector  $\nabla p = \left( \frac{\partial p}{\partial x}, \frac{\partial p}{\partial y}, \frac{\partial p}{\partial z} \right)$ .

Thus,  $\omega_{\nabla p} = \frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz$  as desired.

(c)  $\omega_F^* = F_1 dy dz + F_2 dz dx + F_3 dx dy$ .

Hence,  $d\omega_F^* = \left( \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right) dx dy dz = (\operatorname{div} F) dV$ .

(d) Let the curl of  $\nabla f$  be denoted by  $G$ .

$$\nabla f = \left( \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right).$$

Then,  $G_1 = \left( \frac{\partial(\nabla f)_3}{\partial y} - \frac{\partial(\nabla f)_2}{\partial z} \right) = \left( \frac{\partial^2 f}{\partial z \partial y} - \frac{\partial^2 f}{\partial y \partial z} \right) = 0$ .

$G_2, G_3$  are analogously zero.

(e) Let the curl of  $F$  be denoted by  $G$ .

$$G_1 = \left( \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right), G_2 = \left( \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right), G_3 = \left( \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right).$$

$$\frac{\partial G_1}{\partial x} = \left( \frac{\partial^2 F_3}{\partial y \partial x} - \frac{\partial^2 F_2}{\partial z \partial x} \right), \frac{\partial G_2}{\partial y} = \left( \frac{\partial^2 F_1}{\partial z \partial y} - \frac{\partial^2 F_3}{\partial x \partial y} \right), \frac{\partial G_3}{\partial z} = \left( \frac{\partial^2 F_2}{\partial x \partial z} - \frac{\partial^2 F_1}{\partial y \partial z} \right)$$

Since  $F$  is  $C^2$ , the order of partial derivatives does not matter. Hence, all the terms cancel out to give  $\operatorname{div} G = \frac{\partial G_1}{\partial x} + \frac{\partial G_2}{\partial y} + \frac{\partial G_3}{\partial z} = 0$ .

QED